1.7.2 De Morgan's Laws for Complementation

These laws relate the set operations on union, intersection and complementation.

Let us consider universal set $U=\{0,1,2,3,4,5,6\}$, $A=\{1,3,5\}$ and $B=\{0,3,4,5\}$.

Now, $A \cup B = \{0,1,3,4,5\}$

Then, $(A \cup B)' = \{2,6\}$ .....(1)

Next, $A' = \{0,2,4,6\}$ and $B' = \{1,2,6\}$

Then, $A' \cap B' = \{2,6\}$ .....(2)

From (1) and (2), we get $(A \cup B)' = A' \cap B'$

Thinking Corner

Check whether $A - B = A \cap B'$

Also, $A \cap B = \{3,5\}$,

$(A \cap B)' = \{0,1,2,4,6\}$ .....(3)

$A' = \{0,2,4,6\}$ and $B' = \{1,2,6\}$

$A' \cup B' = \{0,1,2,4,6\}$ .....(4)

Thinking Corner

$(A - B) \cup (B - A') =$ ____

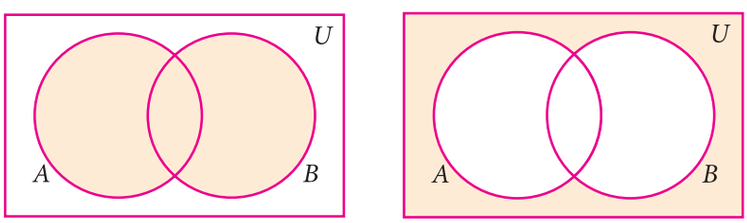
From (3) and (4), we get $(A \cap B)' = A' \cup B'$

De Morgan's laws for complementation : Let 'U' be the universal set containing finite sets A and B. Then (i) $(A \cup B)' = A' \cap B'$ (ii) $(A \cap B)' = A' \cup B'$

Example 1.25

Verify $(A \cup B)' = A' \cap B'$ using Venn diagrams.

Solution



$A \cup B$ $(A \cup B)'$

.... (1)

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